Native hashed ECDL: sampling and recursive audit

2026-10-29 · Development correctness; no performance or security claim

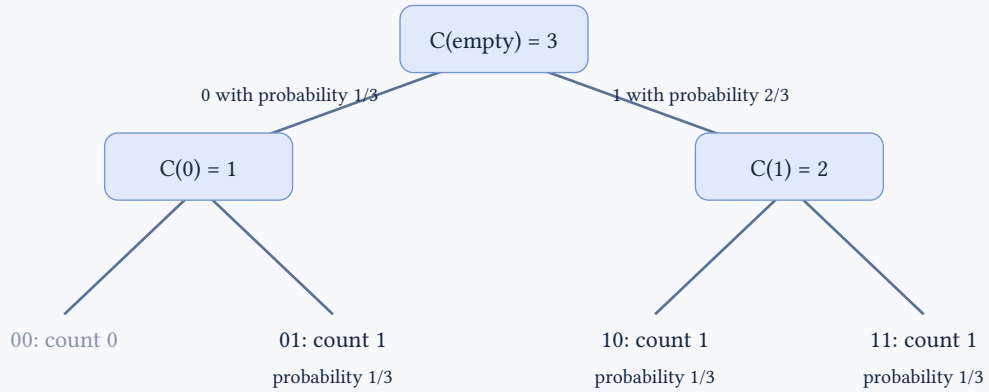
The six supplied approaches already have a native Rust implementation in `crates/hashed-ecdl-tensor`: the relation reduction, exact Boolean counter and deterministic witness finder, Fourier multiplier, exceptional point addition, affine-hashed bases and source parameter guards. Its byte-identical mirror is in cryptanalysis. This change adds the previously absent random sampling path of Bochum Lemma 1 and strengthens forced recursive controls. Coordinate predicates access x and y, placing this reduction outside the generic-group oracle model; that fact alone establishes no speed advantage.

Exact prefix sampling

For current count C and zero-prefix count C0, choose zero with probability C0/C and one with probability (C-C0)/C. Forced branches use no entropy. Pins condition the distribution. Empty languages return `None`; work and entropy errors propagate. One initial contraction plus at most one per unpinned interface bit suffices. Sampling is uniform over language words only for a parsimonious encoding; otherwise auxiliary multiplicities determine weights.

Lemma 1: exact counts determine the next bit

Example predicate x OR y; all three satisfying assignments have one auxiliary completion.



Hide y: the two interface words x=0 and x=1 have weights 1:2, rather than equal probability.

Software controls verify both distributions. General contraction correctness and complexity remain unaudited.

Two focused tests check every draw in a six-element equiprobable fixture, the 1:2 projection weights, pinned and empty interfaces, empty languages, entropy failure and zero work budget. These are exact distribution checks, without statistical tolerances. Existing deterministic relation finding is unchanged.

Both taper levels

Existing forced controls used u=1 and odd Hermite nodes. The new control uses u=2, a=2, L=4 and radix 2, two parallel edges and a degree-4 extension ring at precision 3. Three label layouts and three predicate pairs give nine exact comparisons with direct contraction. Valuations zero and one exercise different coordinates, multiplicities and child precisions. Forced tests bypass source admission: they do not verify admitted general recursion or its asymptotic cost.

Cost interpretation

For the illustrative exponent E=b/m+mb/sqrt(log2(b)), minimizing in continuous m gives m=(log2(b))^(1/4) and E=2b/(log2(b))^(1/4). Its exponent equals rho's b/2 at log2(b)=256, or b=2^256 bits. At b=256, integer m=2 gives about 309.02 against rho's 128. These are illustrative derived exponents, not operation measurements. Setting the constant to one after collapsing the expression instead gives equality at log2(b)=16. Neither convention supplies an actual cryptanalytic crossover.

Evidence and limits

The starting native panel passed 27 tests. New focused controls pass; final commands, outcomes and failures are in `validation.txt`. The full component map, pinned primary-source links and remaining obligations are in `README.md`. Primary review covered Bochum v0.2 §§2–3 and the October 25 OpenAI manuscript, Appendix D.4.1–D.4.6, including immutable creation precision and tapered recovery.

On this 64-bit build, lambda is at most 64 and root_u=lambda/131072 is zero: the public API always selects finite-size fallback. Forced tests bypass admission. General correctness, complexity and space accounting remain independently unaudited. No benchmark, cloud experiment, deployed-curve run, new theorem or Lean proof is claimed. No canonical IC/rho graph changes because no admitted measurement changes. The canonical problem remains `problems/hashed-ecdl-tensor-backend.md`.